13.3 Split Brain Resolver

The problem that the Split Brain Resolver (SBR) solves is how to automatically shut down nodes that become unreachable under certain rules. To set it up, add the following to your configuration:

```
akka.cluster.downinig-provider-class = "akka.cluster.sbr.SplitBrainResolver"
```

Becoming *unreachable* is a relative term that expresses what a node experiences relative to another node. The formation and maintenance of a cluster are done by communication between the nodes using a protocol called *gossip*. Each node sends messages to all the others to check whether they are reachable and responsive. When you combine unreachability with Akka sharding—which distributes actors evenly across the cluster—you can run into the following problem.

13.3.1 An unreachable problem

Imagine that one node becomes unreachable from the other nodes of the cluster. All sharded entities located on that node are replicated to the healthy nodes. But in that case, the entities are duplicated.

To understand this problem better, let’s look at what happens when the cluster splits. By now you may be familiar with the states of a node, which are shown again in figure 13.1.